Lecture- 9.

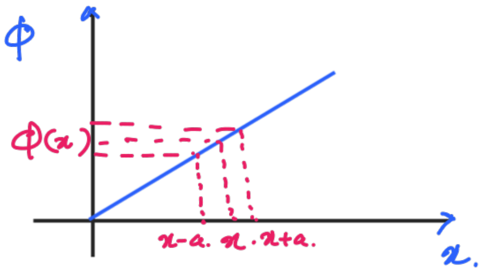
Solving Laplace's Equation in 2D

5th March, 2025

Office: EE514, aneesh@ee.iith.ac.in

Laplace's Eqn in 1D

$$\nabla^2\phi = 0 \Rightarrow \frac{\partial^2\phi}{\partial x^2} = 0 \Rightarrow \phi = mx + c \quad ; \text{ if } c = 0, \ \phi = mx.$$



$\phi(x) \approx \dfrac{\phi(x+a) + \phi(x-a)}{2}$.

$\Rightarrow$ No minimum.

$\Rightarrow$ It can always be an average value.

Laplace's Eqn in 2-D

$\phi(x,y)$

$$\nabla^2\phi = 0 \Rightarrow \frac{\partial^2\phi}{\partial x^2} + \frac{\partial^2\phi}{\partial y^2} = 0 \text{ ------ (1)}$$

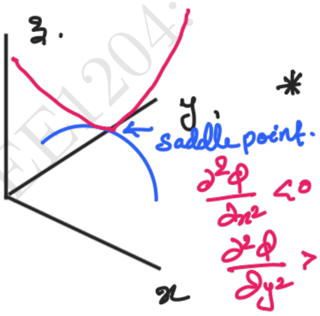
$\Rightarrow \dfrac{\partial^2\phi}{\partial x^2} < 0 \ , \ \dfrac{\partial^2\phi}{\partial y^2} > 0$

oscillating periodic function

Non periodic, exponential function.

OR

$\dfrac{\partial^2\phi}{\partial x^2} > 0 \ , \ \dfrac{\partial^2\phi}{\partial y^2} < 0$.



$\dfrac{\partial^2\phi}{\partial x^2} < 0$

$\dfrac{\partial^2\phi}{\partial y^2} > 0$.

* Recall : If a function 'ϕ' is having oscillatory behaviour $\Rightarrow$ its second derivative is −ve

If it is having exponential characteristics $\Rightarrow$ its second derivative is +ve.

* Laplace's Eqn defines the function which is the saddlepoint at every point (x,y).

General Solution for Eqn (1)

$$\phi(x,y) = X(x)\,Y(y)$$

$$\Rightarrow \frac{\partial^2 (X(x)Y(y))}{\partial x^2} + \frac{\partial^2}{\partial y^2}\left(X(x)\,Y(y)\right) = 0$$

$$\Rightarrow Y(y)\frac{\partial^2 X(x)}{\partial x^2} + X(x)\frac{\partial^2 Y(y)}{\partial y^2} = 0 \text{ ----}$$

$\div\, \phi(x,y)$

$$\Rightarrow \frac{1}{X(x)}\underbrace{\frac{\partial^2 X(x)}{\partial x^2}}_{\text{full derivative in '}x\text{'}} + \frac{1}{Y(y)}\underbrace{\frac{\partial^2 Y(y)}{\partial y^2}}_{\text{full derivative in '}y\text{'}} = 0.$$

$$\Rightarrow \underbrace{\frac{1}{X(x)}\frac{d^2 X(x)}{dx^2}}_{F(x)} + \underbrace{\frac{1}{Y(y)}\frac{d^2 Y(y)}{dy^2}}_{G(y)} = 0 \text{ -----(2)}$$

(Variable separable)

Eqn (2) should be valid for every (x,y).

ie; $F(x) + G(y) = 0$.

Assume at $x=0$ $F(0) = 10$.

$\Rightarrow G(y) = G(0) = -10$

Now if at $x = 1$; $F(1) = 11$

$\Rightarrow F(1) - \overset{\to G(0)}{10} = 0$.

$11 - 10 \neq 0$.

ie; $F(0) = F(x) = 10 \;\forall x$.
$G(0) = G(y) = -10 \;\forall y$.

$\Rightarrow F(x) = \text{Constant}$; $G(y) = \text{Constant}$.

$$\Rightarrow \quad \frac{1}{X}\frac{d^2X(x)}{dx^2} = -\frac{1}{Y}\frac{d^2Y(y)}{dy^2} = +k^2$$

$$\Rightarrow \quad \frac{d^2}{dx^2}X(x) - k^2X = 0 \quad \Rightarrow X = C_1 e^{kx} + C_2 e^{-kx}.$$

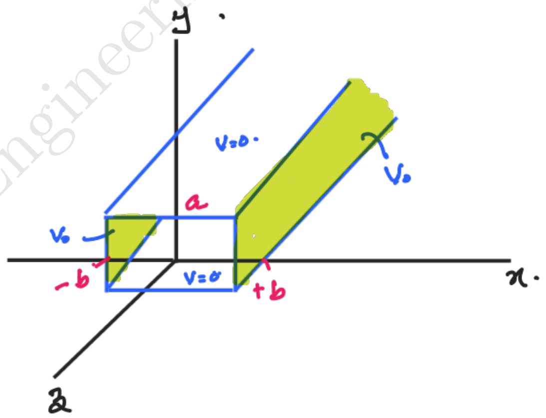
$$\frac{d^2}{dy^2}Y + k^2Y = 0. \quad \Rightarrow Y = C_3 \cos ky + C_4 \sin ky.$$

Soln $\Rightarrow \boxed{\Phi = XY}$.

Cylindrical → Ex: Fiber optic
Oscillatory! ⇒ Bessel functions

Laplace's 2D problem: Example.

Two infinitely-long grounded metal plates, again at $y=0$ & $y=a$, are connected at $x=\pm b$, by metal strips maintained at constant potential V_0. (a thin layer of insulation at each corner prevents them from shorting out). Find the potential inside the resulting rectangular pipe?



Solution:

Configuration is independent of 'z' $\therefore$ Laplace's Eqn $\nabla^2\phi = 0$

$$\Rightarrow \frac{\partial^2\phi}{\partial x^2} + \frac{\partial^2\phi}{\partial y^2} = 0.$$

subject to boundary conditions (B.C):

(i) $\phi = 0$ when $y=0$, $y=a$.

(ii) $\phi = V_0$ when $x = +b$, $x = -b$.

$$\Rightarrow \phi(x,y) = \left(C_1 e^{kx} + C_2 e^{-kx}\right)\left(C_3 \sin ky + C_4 \cos ky\right)$$

Due to the symmetry w.r.t x here

$$\phi(x,y) = \phi(-x,y)$$

$$\left(C_1 e^{kx} + C_2 e^{-kx}\right)\left(C_3 \sin ky + C_4 \cos ky\right) =$$

$$= \left(C_1 e^{-kx} + C_2 e^{kx}\right)\left(C_3 \sin ky + C_4 \cos ky\right)$$

$$\Rightarrow C_1\left[e^{kx} - e^{-kx}\right] = C_2\left[e^{kx} - e^{-kx}\right]$$

$$\Rightarrow C_1 = C_2.$$

$$\Rightarrow C_1' = 2C_1 \Rightarrow C_1' \cosh kx \left(C_3 \sin ky + C_4 \cos ky\right) = \phi(x,y)$$

$$\Rightarrow \cosh kx \left[C_3' \sin ky + C_4' \cos ky\right] = \phi(x,y)$$

$$\Rightarrow \cosh kx \left[C_3' \sin ky + C_4' \cos ky \right] = \phi(x,y)$$

Apply B.C (i) $\phi = 0$ when $y=0$, $y=a$.

$\Rightarrow$ $y=0 \Rightarrow \cosh kx \cdot C_4' = 0.$

$\Rightarrow C_4' = 0.$

$$\Rightarrow \phi(x,y) = C_3' \cosh kx \sin ky.$$

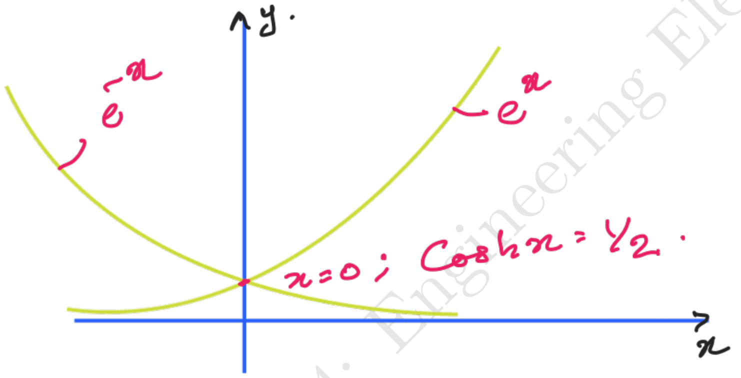
$\phi = 0$; at $y = a$

$$\Rightarrow C_3' \cosh kx \sin ka = 0.$$

; $\cosh kx \neq 0$ $\forall x$. $C_3' \neq 0 \Rightarrow$ As $C_3' = 0 \Rightarrow \phi = 0$ $\forall x, y$.

$\Rightarrow \sin ka = 0.$

$\Rightarrow k = \frac{n\pi}{a}$; $n = 1, 2, \ldots$



$$\Rightarrow \phi(x,y) = C_3' \cosh \frac{n\pi}{a} x \sin \frac{n\pi}{a} y.$$

$\boxed{\phi(x,y) \text{ does not know anything about 'n'.}}$

Now If $\phi_1(x,y)$, $\phi_2(x,y)$ are both solutions of Laplace's equation, then $\phi_1 + \phi_2$ is also solution.

ie, $\left.\begin{array}{l}\nabla^2 \phi_1(x,y) = 0 \\ \nabla^2 \phi_2(x,y) = 0\end{array}\right\} \Rightarrow a\phi_1 + b\phi_2$ will be solution. $a, b \in \mathbb{R}$

ie, Linear Combination will be solution.

$n \neq 0$ as $\phi(x,y)=0$ if $n=0$.

$$\therefore \quad \phi(x,y) = \sum_{n=1}^{\infty} C_n \cosh\left(\frac{n\pi x}{a}\right) \sin\left(\frac{n\pi y}{a}\right)$$ ← Simple linear Combination of Solutions.

Apply B.C at $x=b$, $\phi=V_0$.

$$\Rightarrow \phi(b,y) = \sum_{n=1}^{\infty} \underbrace{C_n \cosh\left(\frac{n\pi b}{a}\right)}_{a_n} \sin\left(\frac{n\pi y}{a}\right) = V_0.$$

$$\Rightarrow \boxed{\sum_{n=1}^{\infty} a_n \sin\left(\frac{n\pi y}{a}\right) = V_0.}$$ → Sine Fourier Series.

$$\frac{1}{2}\int_0^a \cos\left(\frac{n-m}{a}y\right) - \cos\left(\frac{n+m}{a}y\right) dy.$$

$= 0$ if $m \neq n$.
$= \frac{a}{2}$ if $m=n$.

$$\int_0^a \sin\left(\frac{n\pi y}{a}\right)\sin\left(\frac{m\pi y}{a}\right) dy = 0 \quad \text{if } m \neq n.$$
$$= \frac{a}{2} \quad \text{if } m=n.$$

$$\Rightarrow \sum_{n=1}^{\infty} a_n \int_0^a \sin\left(\frac{n\pi y}{a}\right)\sin\left(\frac{m\pi y}{a}\right) dy = \int_0^a V_0 \sin\left(\frac{m\pi y}{a}\right) dy.$$

$$\Rightarrow a_m\left(\frac{\cancel{a}}{2}\right) = -\frac{V_0 a}{m\pi}\cos\left(\frac{m\pi y}{a}\right)\Big|_0^a.$$

$$= \frac{V_0 \cancel{a}}{m\pi}\left[1 - \cos(m\pi)\right]$$

$$\Rightarrow a_m = a_n = \frac{2V_0}{m\pi}\left[1 - \cos(m\pi)\right]$$

$$\Rightarrow \quad a_n = \begin{cases} 0 & \text{if } n = \text{odd} \\ \dfrac{4V_0}{n\pi} & \text{if } n = \text{even} \end{cases}$$

$$a_n = C_n \cosh\left(\frac{n\pi b}{a}\right)$$

$$\Rightarrow \quad C_n = \frac{a_n}{\cosh\left(\frac{n\pi b}{a}\right)}$$

Solution:

$$\Phi(x,y) = \sum_{n=1}^{\infty} C_n \cosh\left(\frac{n\pi x}{a}\right) \sin\left(\frac{n\pi y}{a}\right)$$

$$= \sum_{n=1}^{\infty} a_n \frac{\cosh\left(\frac{n\pi x}{a}\right)}{\cosh\left(\frac{n\pi b}{a}\right)} \sin\left(\frac{n\pi y}{a}\right)$$

$$\boxed{\Phi(x,y) = \frac{4V_0}{\pi} \sum_{n=1,3,5,\ldots}^{\infty} \frac{1}{n} \frac{\cosh\left(\frac{n\pi x}{a}\right)}{\cosh\left(\frac{n\pi b}{a}\right)} \sin\left(\frac{n\pi y}{a}\right)}$$